# Information Security

## Mock 3

Academic Year 2026–2027

**Duration:** 1 h 30

**Instructions.** Answer all questions. Responses must be *concise* and every claim must be *justified*. Calculators and external material are not permitted.

## Part I – Web Privacy and Data Protection

The campus computing centre operates a shared web server reachable at the host `campus.example`, which publishes material for courses and for individual members of the campus. The following facts are known:

- Any student or staff member may upload files to publish a personal page. For instance, a user named Alice can upload HTML and JavaScript and serve it from `http://campus.example/users/alice/`.
- The server runs legacy software and accepts connections over both HTTP and HTTPS.

The staff of course CS123 additionally operate an assignment submission application at `http://campus.example/courses/cs123/submissions`. A student signs in with an email address and password; upon successful authentication the application sets a cookie containing a session identifier, which keeps the student signed in for 24 hours without re-entering credentials. Inspecting the cookie reveals the following attributes:

| Attribute | Value |
|---|---|
| Name | `sessionid` |
| Value | (random session identifier) |
| Path | `/` |
| Domain | `campus.example` |
| SameSite | `Strict` |

**Q1.** While Bob, a CS123 staff member, is using the submission application over `http://campus.example/courses/cs123/submissions`, the attacker Eve passively monitors his network traffic. Can Eve recover the value of the `sessionid` cookie? State your reasoning.

**Q2.** Alice may publish arbitrary content at `http://campus.example/users/alice/`, but holds no further privileges on the server: she cannot modify other pages and cannot read the access logs. Determine whether Alice is able to obtain Bob's `sessionid` cookie. If she is, describe concretely how she would construct her page to capture it; if she is not, explain what prevents her.

**Q3.** The administrators weigh several changes to the way the cookie is issued.

(a) Suppose the `Secure` attribute is added to the cookie. First, state what Bob must do

differently when he connects to the submission application. Then determine whether this change alters either of your two previous answers, and explain why.

(b) Suppose that, instead of `Secure`, the `HttpOnly` attribute is added. Determine whether this change alters either of your two previous answers, and explain why.

**Q4.** For the two independent scenarios below, return to Alice's situation in Question 2 and disregard the attribute changes considered in Question 3.

(a) The submission application is relocated to the host `cs123.campus.example`, and the cookie's `Domain` attribute is set to `cs123.campus.example`. Does your answer to Question 2 still hold? Explain.

(b) Alternatively, the submission application remains at `campus.example` with `Domain=campus.example`, but personal pages are migrated to individual subdomains, so that Alice now publishes at `alice.campus.example`. Does your answer to Question 2 still hold? Explain.

## Part II – Cryptography

**Scenario.** Alice is building a secure messaging application intended to protect confidential corporate documents that she exchanges first with Bob and later with further colleagues. The adversary Eve can observe all network traffic, retain copies of transmitted messages, and on occasion substitute messages of her own.

### A. Block-cipher internals and key establishment

Alice adopts AES-128 as the encryption scheme for her application.

**Q1.** One round of AES applies the MixColumns step, in which each column of the state is multiplied, over the field $\mathrm{GF}(2^8)$, by a fixed matrix. For the matrix $M$ and the input column $c$ given below, compute *only the first byte* of the resulting column. Show every intermediate step of the finite-field arithmetic.

$$M = \begin{bmatrix} \texttt{03} & \texttt{01} & \texttt{01} & \texttt{02} \\ \texttt{02} & \texttt{03} & \texttt{01} & \texttt{01} \\ \texttt{01} & \texttt{02} & \texttt{03} & \texttt{01} \\ \texttt{01} & \texttt{01} & \texttt{02} & \texttt{03} \end{bmatrix}, \qquad c = \begin{bmatrix} \texttt{A3} \\ \texttt{C1} \\ \texttt{5E} \\ \texttt{7B} \end{bmatrix}.$$

**Q2.** Alice and Bob share no secret key in advance. Describe a method by which they can agree on a common secret value over a channel that Eve can fully observe, without that secret being exposed to her. Give the sequence of steps.

### B. Analysis of a fair-decision protocol

Alice and Bob wish to settle a dispute by means of a fair, coin-flip-like decision taken over an insecure network, using only exchanged messages. Alice draws an integer $n$ at random; independently, Bob bets on its parity by announcing *even* or *odd*. Bob wins if his guess matches the parity of $n$; otherwise Alice wins.

**Q3.** Consider the naive protocol below:

- Alice draws a random integer $n$, and Bob independently selects *even* or *odd*.
- Each party transmits its choice in the clear.

- The parity of $n$ is compared with Bob's guess to decide the winner.

Identify the opportunities for cheating that this protocol permits.

**Q4.** To harden the protocol, Alice introduces a symmetric cipher $E$:

- Alice picks a secret key $K$ and sends $C = E_K(n)$ to Bob.
- Bob announces *even* or *odd*.
- Alice then discloses $(n, K)$ so that Bob can recompute and verify $C$.

(a) Identify any remaining opportunities for cheating, and justify your answer.

(b) Could the symmetric cipher be replaced by a public-key encryption scheme? Would such a substitution make the protocol secure? Justify your answer.

**Q5.** Alice and Bob now consider using a cryptographic hash function $H$ in place of encryption.

(a) Design a protocol that lets them decide the winner fairly. Specify the messages exchanged and the verification each party performs.

(b) Define the *collision-resistance* property of a cryptographic hash function. What happens to your protocol if $H$ fails to be collision-resistant? Explain.

(c) Define the *preimage-resistance* property of a cryptographic hash function. What happens to your protocol if $H$ fails to be preimage-resistant? Explain.

— End of questions —